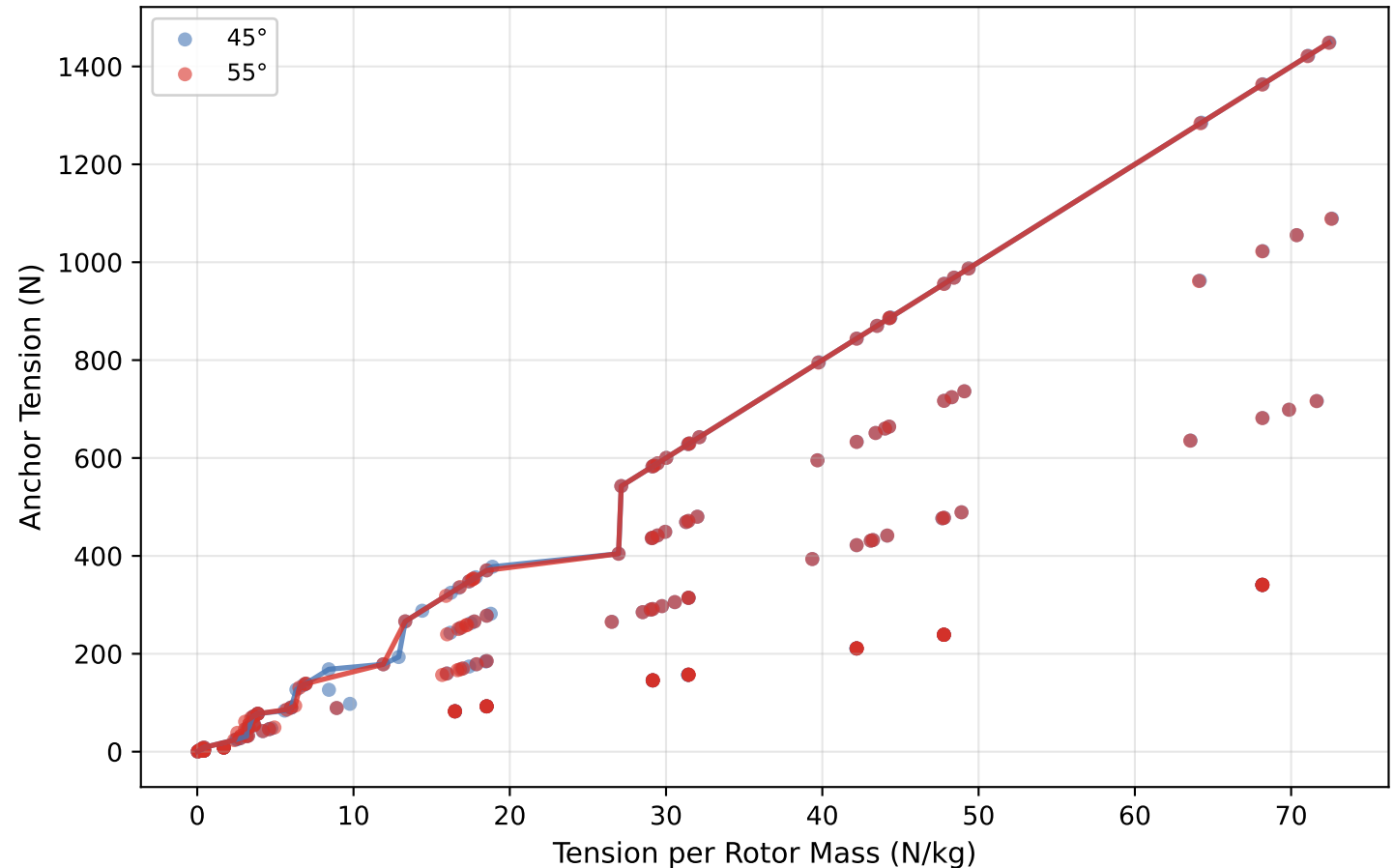


## Pareto: Tension vs Mass Efficiency by Line Elevation BEM v2.1 — All (R,N,profile) Combinations



*IMPLICATIONS: Line elevation has a small but consistent effect. 55° (steeper) pushes the Pareto frontier slightly right — better mass efficiency at the same absolute tension — because the through-disk velocity increases with  $\sin(\text{elevation})$ . However, the effect is subtle: the 45° and 55° clusters overlap heavily. The practical difference (~5-10% in N/kg) is within the noise of manufacturing tolerances. Fly at whichever angle gives best clearance and line management. Data: BEM v2.1, 384 configs.*